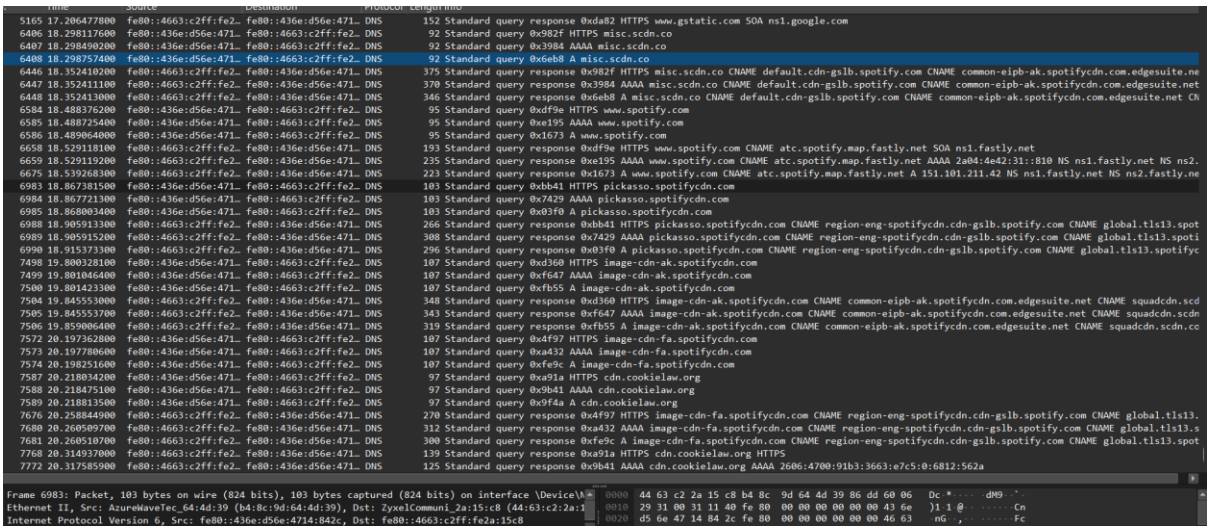
Day-1

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.



Wireshark was used to capture network traffic while opening www.spotify.com in a web browser.

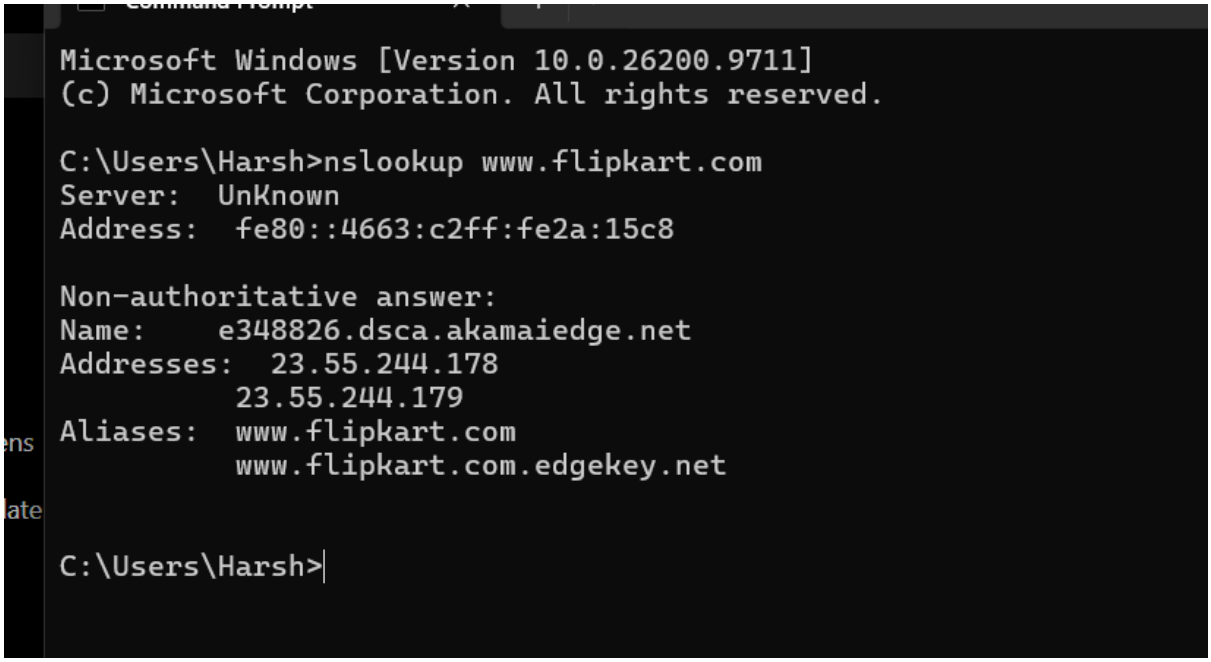
1. DNS Packets

No.	Packet No.	Protocol	Details
1	6584	DNS	Query for www.spotify.com to resolve its IP address.
2	6675	DNS	Response containing the IPv4 address 151.101.211.42 for www.spotify.com.

2. HTTPS-Related Packets

No.	Packet No.	Protocol	Details
3	To be identified	TLS	Encrypted communication between the browser and Spotify's servers.
4	To be identified	TLS	Encrypted application data exchanged during website loading.

2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.



Parameter	Value
Domain Name	www.flipkart.com
DNS Server	Unknown
IP Address 1	23.55.244.178
IP Address 2	23.55.244.179
Alias	www.flipkart.com.edgekey.net

Explanation

The nslookup command successfully resolved www.flipkart.com to the IP addresses 23.55.244.178 and 23.55.244.179. This means the DNS system translated the domain name into IP addresses that the computer can use to connect to the website.

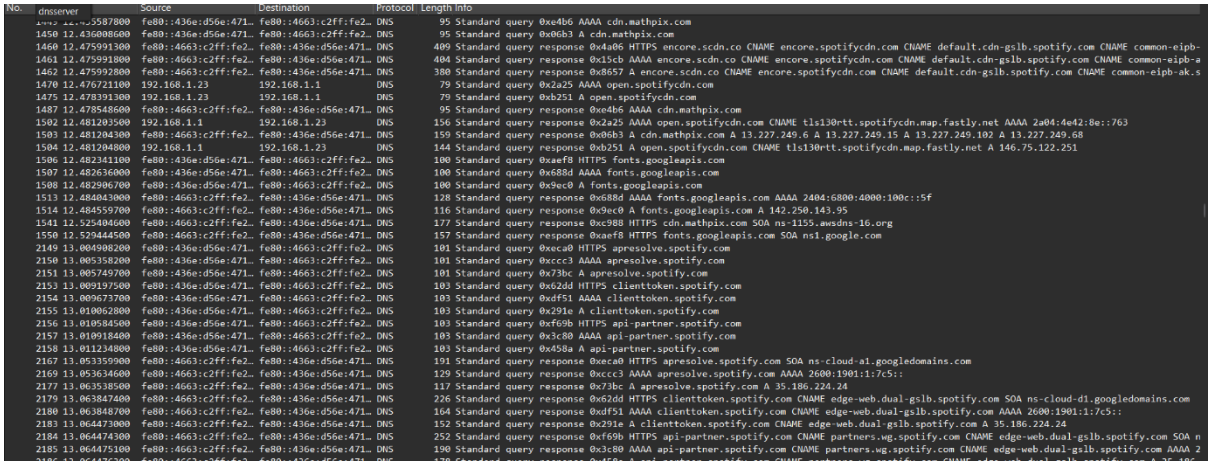
3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Key Difference	HTTP	HTTPS
1. Security	Data is transmitted without encryption.	Data is encrypted using TLS to protect communication.

2. Port Number	Uses port 80 by default.	Uses port 443 by default.
3. Data Protection	Data can be intercepted and read more easily.	Encryption helps prevent attackers from reading or modifying data in transit.

Real-world example: Flipkart uses HTTPS to protect communication between users' browsers and its servers when they log in, browse products, or make payments.

4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.



Wireshark was opened and the DNS display filter (dns) was applied. A website was accessed in the browser to observe DNS queries and responses.

Observations

Step	Protocol Activity	Observation
1	DNS Query (A)	The computer requests the IPv4 address of a domain name.
2	DNS Query (AAAA)	The computer requests the IPv6 address of the domain name.
3	DNS Response (A)	The DNS server returns the IPv4 address if available.
4	DNS Response (AAAA)	The DNS server returns the IPv6 address if available.
5	CNAME Response	Some domains return an alias pointing to another domain, such as a CDN hostname.

Description of the Screenshot

The screenshot shows DNS queries and responses for Spotify-related domains, including open.spotify.com, api-partner.spotify.com, and clienttoken.spotify.com. Both A and AAAA queries are visible, and some responses contain CNAME records pointing to other Spotify or CDN hostnames.

DNS translates domain names into IP addresses so the computer can connect to websites and services. The capture demonstrates how a browser may perform multiple DNS lookups for a website's main domain and its supporting services.

5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Protocol: DHCP (Dynamic Host Configuration Protocol)

Scenario: When I visit a café and connect my phone to its Wi-Fi to open Instagram, the café's router uses DHCP to automatically assign my phone an IP address, subnet mask, default gateway, and DNS server address. This allows my phone to communicate with the local network and access the internet without manually configuring network settings.